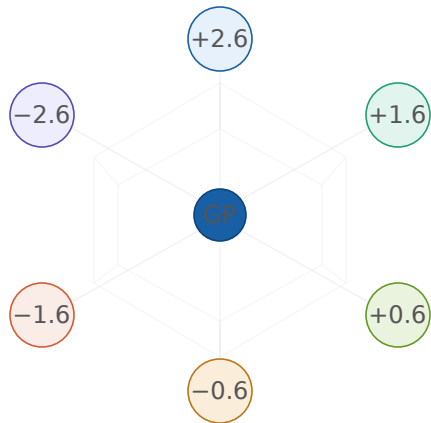




600-cell lattice

120 vertices · $z=12$ · H_4 symmetry
6 distinct eigenvalues (diameter 5)

six distinct eigenvalues

$$\varphi = (1+\sqrt{5})/2 = 1.618$$

colour	λ	value	SM gen.?
blue	$+(1+\varphi)$	+2.618	1st gen. (open)
teal	$+\varphi$	+1.618	1st gen. (open)
green	$+(\varphi-1)$	+0.618	2nd gen. (open)
orange	$-(\varphi-1)$	-0.618	2nd gen. (open)
red	$-\varphi$	-1.618	3rd gen. (open)
purple	$-(1+\varphi)$	-2.618	3rd gen. (open)

$$[a_k, a^\dagger(k')] = \delta_{(kk')}$$

from eigenvector orthonormality